HWA CHONG INSTITUTION (COLLEGE)
JC2 H1 PRELIMINARY EXAMINATIONS 2008
BIOLOGY 8875 – MARK SCHEME

Paper 1 (MCQ)

1.	**A**	6.	**D**	11.	**A**	16.	**C**	21.	**A**	26.	**D**
2.	**C**	7.	**A**	12.	**A**	17.	**B**	22.	**D**	27.	**A**
3.	**C**	8.	**B**	13.	**D**	18.	**C**	23.	**B**	28.	**C**
4.	**D**	9.	**B**	14.	**A**	19.	**D**	24.	**C**	29.	**A**
5.	**C**	10.	**B**	15.	**C**	20.	**C**	25.	**A**	30.	**A**

Paper 2 (CORE)
Section A: Structured Answers

QUESTION 1

(a)(i) State one similarity and two differences in the structure of glycogen and triglyceride. [2]

Similarity : both are made up of **carbon, oxygen and hydrogen**
Difference : *(any 1)*

	Glycogen	Triglyceride
1.	monomers are made up of **glucose**	monomers are made up of **glycerol & fatty acid chains**
2.	bonds linking up the monomers are **glycosidic linkages**	bonds linking up the monomers are **ester linkages**

(ii) Which metabolic fuel is a better form of energy storage in the human body? Give a reason for your answer. [2]

triglyceride

(any 1)

reason 1:
contains a **greater number of carbon atoms per gram**
a larger amount of energy can be stored
more C—C bonds can be broken down to release energy

reason 2:
presence of **many non-polar C—H bonds in hydrocarbon chains of fatty acids / triglyceride is hydrophobic**
organisms **do not carry extra weight of water of hydration / lighter to carry triglycerides**

reason 3:
presence of **many non-polar C—H bonds in hydrocarbon chains of fatty acids / triglyceride is hydrophobic**
therefore can be **stored in large quantities without affecting osmotic potential**

(b) Describe how the hydrogen released during glycolysis and the Krebs cycle results in the production of ATP. [2]

hydrogen released are used **to reduce; coenzymes NAD^+ & FAD^+**
$NADH$ and $FADH_2$ will be **reoxidized** through transfer of their electrons and protons **to the electron transport chain** at the **inner membrane of the mitochondrion**
protons are pumped from the mitochondrial matrix **into the intermembranal space generating an electrochemical gradient**
re-entry of the protons to into the matrix provides the energy to **generate ATP** at the **ATP synthase**
Ref to **chemiosmosis**

(c)(i) Using the information from **Fig. 1.3**, calculate the number of ATP molecules produced from the complete oxidation of a 16-carbon fatty acid. Show your answer in the space provided below. [2]

16-C fatty acid = **8 x 2-C chains**
1 (2-C) chain produces **12 ATP molecules**
8 (2-C) chains produce **12 x 8;** = **96 ATP molecules**

(ii) Suggest why fatty acids can only be respired under aerobic conditions. [2]

fatty acids enter aerobic respiration by being **converted to acetyl coA**
Ref to **link reaction**
occurs at the mitochondrial matrix
in order to be completed oxidized, **oxygen is needed**

QUESTION 2

(a)(i) Identify **DNA Type I** and **II**. [1]

DNA Type I : bacterial chromosome
DNA Type II : bacterial plasmid

(ii) Explain your answers to **(a)(i)** and state one function of **DNA Type II**. [3]

bacterial plasmid contains much **fewer genes**, therefore smaller in size
much **lower molecular weight of 300 kb**
each bacterial cell only contain **1 copy of chromosome**
however, **plasmids can be present in several copies**
plasmids of bacteria contain genes that **confers antibiotic resistance** / enable bacteria to **use a specific nutrient** / can be used as **cloning vectors in research**

(b) Discuss the differences in the genes present in **DNA Type I** and **Type II**. [2]

DNA Type I contains **essential genes**
necessary for the bacteria's survival under normal conditions
DNA Type II contains **genes that are not essential**
useful under specific / stressful environmental conditions

(c) With reference to the **Fig. 2.1**, explain the effect of a single base substitution at the operator region on the *lac* operon. [2]

repressor coded by *lacI* will not be able to recognise and **bind** to the **mutated operator**
the ***lac* operon will not be switched off**
even in the **presence of glucose / absence of lactose**
***lac Z* / structural genes** will **continuously be expressed**
producing β-galactosidase / enzymes

(d)(i) State any difference there might be in the growth rate between the two tubes. [1]

no difference initially; thereafter higher growth rate in the tube with lactose

(ii) Explain your answer to **(d)(i)**. [3]

this is because at **high glucose concentration, cAMP concentration falls**
there is thus **no binding between cAMP and CAP / CRP**
CAP / CRP assumes an **inactive shape, detaching from *lac* operon**
transcription of the *lac* operon proceeds at only **a low level, even in the presence of lactose**
after **glucose is fully utilized, *lac* operon is switched on**
and **growth rate in tube with lactose increases**

QUESTION 3

```
England         AAAGCAGAAGAAATAGATGCAAAGAAGAAGAGTTCAA
Japan           --GCAT----------------C----------AATT
South Africa    ------C-------------------TA----------
India           GG-CG--C-----------GGG-GG-------CGGG
China           TTT---T-------------TT----T-----GGTT
United States   ------C-------------T---------------
Philippines     TTT---T-------------TC----T-----GGTT
New Zealand     ------C---------------C-TA----------
Russia          --GCAT--------------CC----------A-TT
Yemen           ------------------------------C------
```

(a)(i) Draw a box in **Fig. 3.1** to include the entire region that most likely codes for an important structure in the mitochondrion. [1]

(ii) Explain your answer to **(a)(i)**. [2]

consensus sequence / by comparing homologous sections

where **DNA sequence is highly conserved / identical**

due to **absence of nucleotide substitution**

as region codes for an **important part of a protein / protein** that contributes to **normal functioning of mitochondrion**

(b) With reference to **Fig. 3.1**, identify the country of subspecies which is most closely related to the mouse found in England. Give a reason for your answer. [2]

Yemen

due to **single base substitution of sequence**

from G to C

(c) Explain how the large number of mouse subspecies might have evolved. [2]

varying habitats / environmental conditions / different selection pressures in separate populations in different parts of the world (Reject: geographic isolation)
differential survival rate / some more suited to the environment than others
beneficial allele / gene passed on to offspring
no interbreeding / gene flow / reproductive isolation
populations **unable to produce fertile offspring**
(Accept: mutation results in genetic differences

QUESTION 4

(a) Describe how the SGP serves to increase knowledge of the biology of Atlantic salmon. [3]

generate genomic map
to **identify and analyse genes**
as well as study **patterns of inheritance of genes**
and **evolutionary relationships with other organisms**
assist in **wildlife management**
assist in the **construction physical** and **linkage maps**

(b) With reference to **Fig. 4.1**, describe the features of the multiple cloning site, MCS. [2]

consists of a **variety of restriction sites**
such as ***BSSH* II, *Kpn* I and *Sac* I**
that **occur once** in the vector
found **within selection marker *lac* Z**
for **cloning of gene of interest** into vector

(c)(i) It was found that the recombinant plasmid with sGH gene inserted yielded no polypeptide. With reference to **Fig. 4.1**, state and explain a reason for this. [2]

sGH gene inserted in the **wrong orientation**
though **a protein is expressed under control of T3 promoter**
sequence of gene is reversed
3D conformation of protein is different from original

OR

since ***Sac* I restriction site is within / near T3 promoter**
sGH gene inserted **disrupts promoter sequence**
no expression of gene
no polypeptide produced

(ii) Suggest a solution for the problem in **(c)(i)**. [1]

use **2 restriction enzymes** to clone

OR

use **restriction site which is away from T3 promoter** to clone

OR

construct transgene with promoter in front of sGH gene

(d)(i) With reference to **Fig. 4.2**, state the one selection pressure faced by transgenic salmon. [1]

offspring viability / mating advantage

(ii) Using your answer to **(d)(i)**, explain how the named selection pressure may or may not lead to salmon extinction. [2]

Offspring viability:
from **0.7 to 1.0 of offspring viability**, **increase in offspring viability** leads to **increase in time to extinction** from **35 generations to more than 150 generations**
from **0.4 to 0.7 of offspring viability**, **increase in offspring viability** leads to **decrease in time to extinction** from **125 generations to more than 35 generations**

Mating advantage:
as mating advantage decreases, the minimum time to extinction increases for example, minimum time to extinction for a **mating advantage of 4.7** is about **35 generations** as compared to minimum time to extinction of **more than 150 generation**s for a **mating advantage of 1.4**

Paper 2 (CORE)
Section B: Essay Answers

QUESTION 5

(a) Describe the structural features of a named membranous organelle that is involved in ATP synthesis in photosynthesis. [6]

1. **chloroplast**
2. **large / long / elongated / lens-shaped / 5 – 10 μm in length**
3. enclosed by a <u>**double membrane**</u> **/** <u>**inner and outer membrane**</u>
4. **inner membrane is selectively permeable** containing protein channels
5. **outer membrane is permeable** to most ions and molecules
6. contain **gel-like aqueous matrix known as** <u>**stroma**</u>
7. **starch grains / ribosomes / circular DNA / enzymes of Calvin cycle present** *(any 1)*
8. flattened, disc-like sacs known as <u>**thylakoids**</u> **suspended in stroma**
9. arranged in **stacks known as** <u>**grana**</u>
10. which are **joined by** <u>**intergranal lamellae**</u>
11. **thylakoid membrane**
12. contains **photosystems**

12a. **photosynthetic pigments** e.g. chlorophyll and carotenoids are present

13. **electron carriers** / **electron transport chains** are also embedded in the membrane
14. **$NADP^+$ reductase** present
15. **ATP synthase complex / stalked particles** also present
16. **fluid-filled interior space** known as <u>**thylakoid lumen / space**</u>

(b) Compare the two main pathways in which an excited electron can undergo during the light-dependent stage of photosynthesis. [6]

Differences:

Features	**Non-cyclic photophosphorylation**	**Cyclic photophosphorylation**
1. **Photosystems involved**	involves both **PSII** and **PSI**	involves **only PSI**
2. **First electron donor (source of electrons)** 2a.	from **water** **photolysis of water occurs**	from **P700 in PSI** no need water **photolysis of water does not occur**
3. **Electron transport** 3a.	involves **plastoquinone, cytochrome complex, plastocyanin** and **ferredoxin** involves **two electron transport chain**	involves **ferredoxin**, **cytochrome complex** and **plastocyanin only** involves **only one electron transport chain**
4. **Pathway of electron**	**non-cyclic**, electron does not return to same molecule	**cyclic**, electron returns to same molecule
5. **Final electron acceptor**	**$NADP^{\pm}$**	**P700 in PSI**
6. **Generation of proton gradient**	proton gradient generated by H^+ released during **photolysis of water** , and **H^+ that are actively pumped** from the stroma, across the thylakoid membrane**, into the thylakoid lumen by energy released during electron flow**	proton gradient is generated by **H^+ that are actively pumped** from the stroma, across the thylakoid membrane, **into the thylakoid lumen by energy released during electron flow**
7. **Products** 7a.	**ATP** & **NADPH produced** **by-product O_2 produced**	**ATP only** **no O_2 produced**

Similarities:

8. both pathways **occur at the thylakoid membran**
9. both pathways involve **PSI**
10. both involve the **synthesis of ATP**
11. both pathways uses **energy released during electron flow** to actively **pump H^+** from the stroma, across the thylakoid membrane, **into the thylakoid lumen to produce a proton gradient** which drives ATP synthesis **/** involves **chemiosmosis**

(c) With reference to the term 'limiting factor', discuss how two named factors have effects on photosynthesis. [8]

1. rate of photosynthesis, which consists of a series of reactions, is **limited by the slowest reaction in the series**
2. and when the rate of photosynthesis is **limited by more than one factors**
3. the **rate is limited by the factor which is in the shortest supply**
4. it **determines the rate of reaction**

(any 2 named factors)

5. **light intensity**
6. directly affects **light-dependent stage** of photosynthesis
7. at **low light intensities**, **light intensity is the limiting factor**
8. **rate of photosynthesis increases proportionally with increasing light intensity**
9. at **high light intensities**, **rate of photosynthesis decreases**
10. **some other factor becomes the limiting factor**

11. **further increase in light intensity** results in the **light saturation point being reached**
12. beyond this point, **further increase in light intensity will have no effect on the rate of photosynthesis**
13. **light intensity is no longer a limiting factor**
14. **rate of photosynthesis is maximum at this point**

15. **<u>carbon dioxide</u>**
16. a **major limiting factor** since atmospheric CO_2 concentration is low
17. CO_2 is **needed in the light-dependent stage / Calvin cycle**
18. it is **reduced to carbohydrate / sugar**
19. it is **fixed by combining with ribulose-1,5-bisphosphate (RuBP)**
20. **increase in CO_2 concentration results in a proportional increase in the rate of photosynthesis**
21. **<u>temperature</u>**
22. affects **mainly light-independent stage** and, to a certain extent, the light-dependent stage**;**
23. are **enzyme-controlled**
24. **increase in temperature, increases rate of photosynthesis**
25. **rate of photosynthesis doubles for each rise of 10^{o}C** until optimum temperature is reached
26. **rate of photosynthesis decreases at higher temperature**
27. as **enzymes start to denature**

28. **<u>oxygen</u>**
29. **competes with CO_2**
30. for the **active site of the enzyme RuBP carboxylase oxygenase / Rubisco**
31. during the **light-independent stage / Calvin cycle**
32. **high concentration of oxygen decreases the rate of photosynthesis**

QUESTION 6

(a) Describe the structural features of two named nucleic acids involved in protein synthesis. [8]

<u>DNA / deoxyribonucleic acid/ deoxyribose nucleic acid</u>

1a. there are **four types** of **deoxyribonucleotides/ deoxyribose nucleotides**
1b. composed of **Adenine**, **thymine**, **cytosine** and **guanine**
3. nucleotides are linked together by **phosphodiester bond**
4. **DNA molecule is double-stranded**
5. 2 strands held together by **hydrogen bonds** between **complementary bases**/ **2 hydrogen bonds** between **A and T** and **3 hydrogen bonds** between **C and G**
6. bases are found between the 2 sugar-phosphate backbone
7. bases are **stacked** above one another with a distance of **0.34 nm**
8. there is **hydrophobic interaction** between the stacked bases, which stabilizes the molecule
9. the two polynucleotide strands are **anti-parallel** / One polynucleotide strand runs in the **3' → 5' direction** while the other polynucleotide strand runs in the **5' → 3' direction**
10. **DNA molecule adopts a helical structure**
11. there are **10 nucleotides per turn** of the helix, with a length of **3.4 nm**
12. the **sugar-phosphate backbone** is found on the **outside of the helix**
13. the **hydrophobic nitrogenous bases** are found in the **core of the helix**
14. **major** and **minor grooves** are found on the DNA double helix

<u>mRNA</u>

1a. **four types** of **ribonucleotides**
1b. composed of **adenine**, **uracil**, **cytosine** and **guanine**
2. held together by **phosphodiester bond**
3. RNA molecule is **single-stranded**
3a. RNA molecule has a 3' hydroxyl end and a 5' phosphate end
4. has a **5' guanine cap**
5. has **3' poly-A tail**
6. **exons are joined togther** in the coding region
6a. contains a coding region
7. has 5' and 3' untranslated region
8. forms fairly random coil

<u>tRNA</u>

1a. **four types** of **ribonucleotides**
1b. composed of **adenine**, **uracil**, **cytosine** and **guanine**
2. held together by **phosphodiester bond**
3. RNA molecule is **single-stranded**
4. its **2D structure** is in the form of a **cloverleaf**.
4a. formed by complementary base-pairing
5. **one of the loops contains the anticodon**
6. its **3D structure** is a **compact L-shaped structure**
6a. maintained by hydrogen bonds
7. has a **3' CCA stem** for the **attachment of amino acid**

(b) Compare the two processes that have to take place for protein synthesis. [8]

Similarities:

1. both processes **require a template**
2. both processes are **catalyzed by enzymes**
3. both processes involve **complementary base-pairing**

Differences:

Feature	**Transcription**	**Translation**
Location	1a. **in nucleus**	1b. **ribosomes;**
Involvement of ribosome	2a. **not involved**	2b. **ribosome is the binding site for mRNA** and **amino-acid tRNA complex**
Template	3a. **DNA template/coding strand**	3b. **mRNA**
Enzyme	4a. **RNA polymerase catalyses formation of phosphodiester bond** between adjacent ribonucleotides	4b. **peptidyl transferase** catalyzes the **formation of peptide bond**
Bond between basic units	5a. **bond is formed** between the **phosphate group of one nucleotide** and **carbon atom 3 of ribose** of the next nucleotide	5b. **bond is formed** between **carboxyl group of one amino acid** and **amino group of the next amino acid**
Reading of genetic message	6a. **RNA polymerase moves along DNA template/ coding strand**	6b. **ribosome moves along mRNA**

Involvement of tRNA	7a. **not involved**	7b. **tRNA carries amino acids** to ribosomes
Raw material(s)	8a. **free ribonucleotides**	8b. **amino acids attached to tRNA**
Rroduct(s)	9a. **mRNA, rRNA and tRNA**	9b. **polypeptide**
Fate of product(s)	10a. products **exit nucleus** and **migrate to the cytoplasm**	10b. **polypeptide chain remain in the cytoplasm** or **secreted out of the cell**

(c) Discuss two named factors that would affect the rate of protein synthesis. [4]

(any 2 named factors)

named factor 1: stability of mRNA
mRNA that is **more stable**
enables **translation to proceed for a longer period of time**
more proteins can thus be produced per unit time

named factor 2: presence of internal ribosome entry site (IRES)
this allows translation to take place even when the **cap binding complex, eIF-4E**
is not able to **bind to the 5' cap**
increasing the rate of translation

named factor 3: presence of upstream open reading frames (uORFs)
uORFs **decrease translation of downstream gene**
ribosome **translate the uORF** and **dissociate from the mRNA**
before it reaches the coding sequence

named factor 4: presence of translational repressors
translational repressors **decrease the rate of translation**
by **preventing communication between 5' cap and 3' tail**
which is **required for efficient translation**